

Answers

1) a) 1440° b) 2340° c) 3240°

2) a) 128.6° b) 150°

3) a) 5 sides b) 12 sides c) 19 sides

4) equal interior angles, equal exterior angles, and equal sides

5)

# of Sides	Interior Angle Sum	Measure of One Interior Angle (regular polygon)	Sum of Exterior Angles	Measure of One Exterior Angle (regular polygon)
n	$180(n - 2)$	$\frac{180(n - 2)}{n}$	360	$\frac{360}{n}$
14	2160	154.3	360	25.7
24	3960	165	360	15
17	2700	158.8	360	21.2
8	1080°	135	360	45
7	900°	128.6	360	51.4
30	5040°	168	360	12
11	1620°	147.3	360	32.7
12	1800	150°	360	30
6	720	120°	360	60
15	2340	156°	360	24
36	6120	170	360	10°
50	8640	172.8	360	7.2°
4	360	90	360	90°
72	12600	175	360	5°

6) a) 120° b) Answer may vary. The angles don't change.

7) a) 144° b) 157.5° c) 162° d) $\frac{180(n-2)}{n}$

8) No; the sum is 360 degrees for all convex polygons.

9) 36°

10) 33

11) 36